

Social Media and Sentiment Analysis Project Guide

1. Introduction

Social media platforms like Twitter and Instagram contain a large amount of user opinions. Sentiment Analysis is a Machine Learning technique used to determine whether a piece of text is positive, negative, or neutral.

2. Objective

- 1 Understand sentiment analysis concepts
- 2 Collect and preprocess social media data
- 3 Train a machine learning model
- 4 Analyze sentiments of posts or comments

3. Tools & Technologies

- 1 Python
- 2 Libraries: pandas, numpy, sklearn, nltk, matplotlib
- 3 API access (Twitter API or dataset for Instagram)
- 4 Jupyter Notebook or VS Code

4. Project Workflow

- 1 Data Collection
- 2 Data Cleaning
- 3 Text Preprocessing
- 4 Feature Extraction
- 5 Model Training
- 6 Evaluation
- 7 Visualization

5. Data Collection

Twitter data can be collected using APIs or public datasets. Instagram data is usually collected using datasets due to API restrictions.

6. Text Preprocessing

- 1 Convert text to lowercase
- 2 Remove stopwords
- 3 Remove punctuation
- 4 Tokenization
- 5 Stemming or Lemmatization

7. Machine Learning Models

- 1 Naïve Bayes
- 2 Logistic Regression
- 3 Support Vector Machine (SVM)
- 4 Deep Learning (LSTM)

8. Implementation Steps

- 1 Import libraries
- 2 Load dataset
- 3 Clean data
- 4 Convert text into numerical features (TF-IDF)
- 5 Train model
- 6 Test accuracy
- 7 Predict sentiment

9. Evaluation Metrics

- 1 Accuracy
- 2 Precision
- 3 Recall
- 4 F1 Score

10. Output

The output of the project will classify text into Positive, Negative, or Neutral sentiment and can also show graphs for analysis.

11. Conclusion

Sentiment analysis helps businesses and researchers understand user opinions. This project provides practical experience in Machine Learning and NLP.